

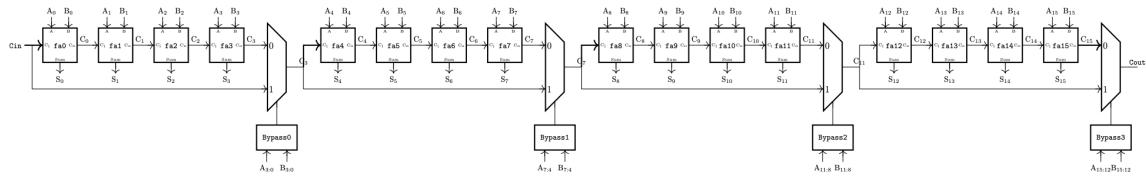
EE5311 Digital IC Design Tutorial 8: 16-bit Carry Bypass Adder

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Assignment Questions

EE5311 Digital IC Design: Assignment 8 Carry bypass adder – schematic and layout extracted simulation

1. Schematics and symbols for `full adder`, `xor`, `mux2.1`, `nand4`, `inverter` gates are in `/cad/share/tutorial.8` directory. Copy over these files to your working directory. Construct a 16-bit carry bypass adder using these gates as shown below:



- (a) Trace the critical path in the above schematic and come up with an input pattern that activates the critical path.
 - (b) Create a SPICE stimulus for the pattern and measure the critical path delay in `ngspice`.
2. Firstly run the `update_std_cells.sh` command to get the required standard cell layouts in KLayout. Close and Open KLayout to get the new standard cells. Draw the layout for the 16-bit adder using the cell layouts provided in the updated `custom.sky130` library.
 - (a) Ensure that the layout is LVS/DRC clean
 - (b) Measure the post-layout extracted delay of the critical path

Figure 1: Assignment Questions for 16-bit Carry Bypass Adder

1. 16-bit Carry Bypass Adder Schematic

This section presents the schematic implementation of the 16-bit carry bypass adder using full adders, XOR gates, MUX21, NAND4, and inverter gates as building blocks.

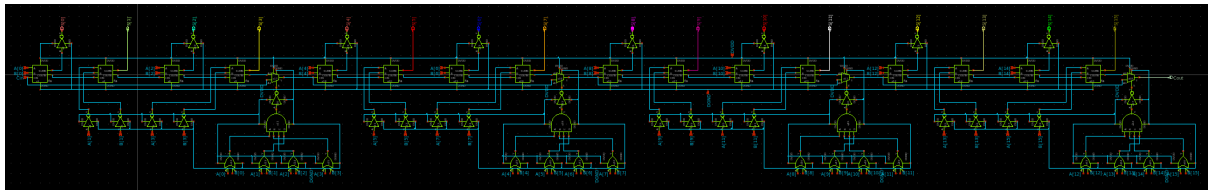


Figure 2: 16-bit Carry Bypass Adder Schematic

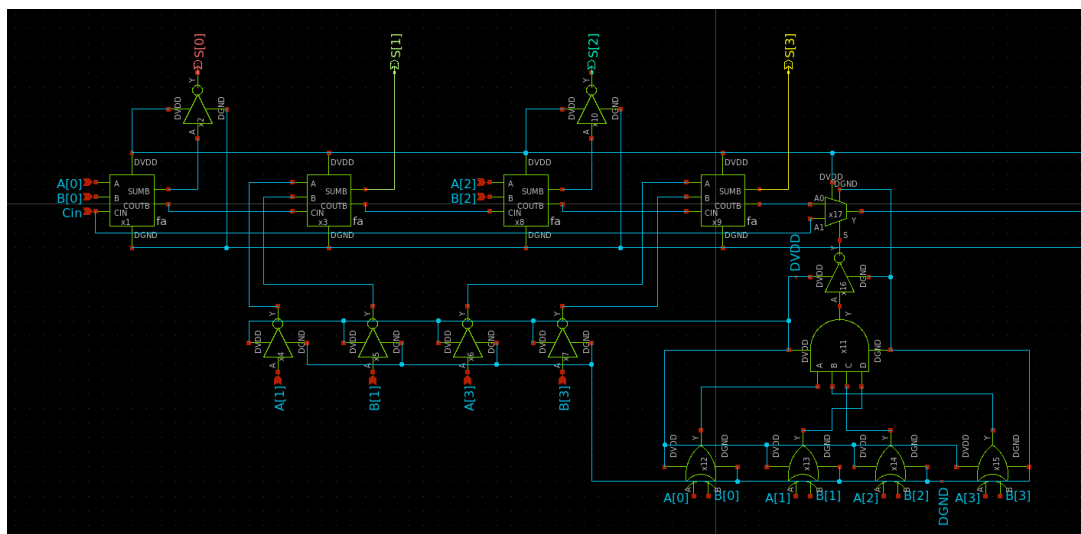


Figure 3: 4-bit Carry Bypass Adder Building Block

2. Critical Path Analysis

The critical path analysis identifies the longest delay path in the 16-bit carry bypass adder architecture.

2.1 Critical Path Identification

The input pattern is given in such a way that the first Full Adder block is in generate mode and rest all will be in the propagation mode of operation and in first half cycle of A_{in} the sum will get to go to 0 as first full adder is in generate mode of operation and as sizes of pmos are less so it will take much time if sum rises from 0 to 1 so here t_2 is the time for which the sum will get max delay and that is the critical path for the Carry bypass adder here. As till then the carry is generated. The carry will get rippled through the First four full adders and reaches the final stage through the NAND and the MUX, this makes the critical path of the sum is more than the carry path delay.

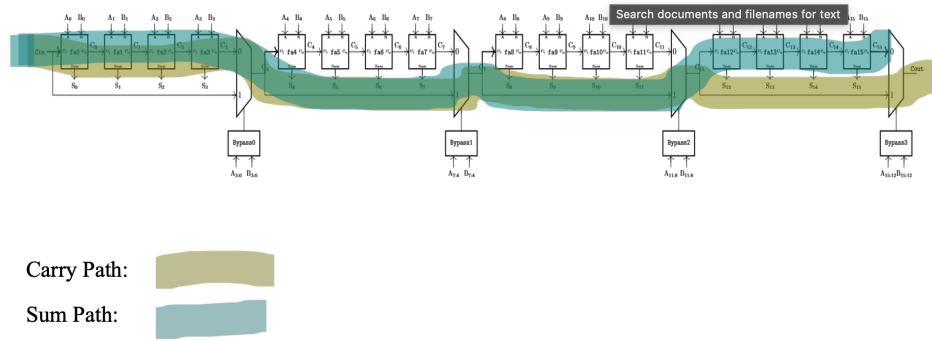


Figure 4: Critical Path Analysis showing Carry Path and Sum Path

3. SPICE Stimulus Code for Critical Path

3.1 Input Pattern for Critical Path Activation

The input pattern is designed such that:

- First Full Adder block operates in generate mode
- Remaining blocks operate in propagation mode
- This creates the worst-case delay scenario

Listing 1: SPICE Stimulus Code

```
.control
tran 0.1n 50n
plot v(in) v(Cout) v(S15)
meas tran tcarry trig v(in) val = 0.9 rise=1 targ v(Cout) val=0.9 rise=1
meas tran thl trig v(in) val = 0.9 rise=1 targ v(S15) val=0.9 fall=1
meas tran tlh trig v(in) val = 0.9 fall=1 targ v(S15) val=0.9 rise=1
let delay = ($&thl + $&tlh) /2
echo S15 Delay : $&delay
echo Carry Delay : $&tcarry
end
.endc
```

4. Pre-layout Measurements

4.1 Pre-layout Simulation Results

The pre-layout simulation provides the following critical path delays:

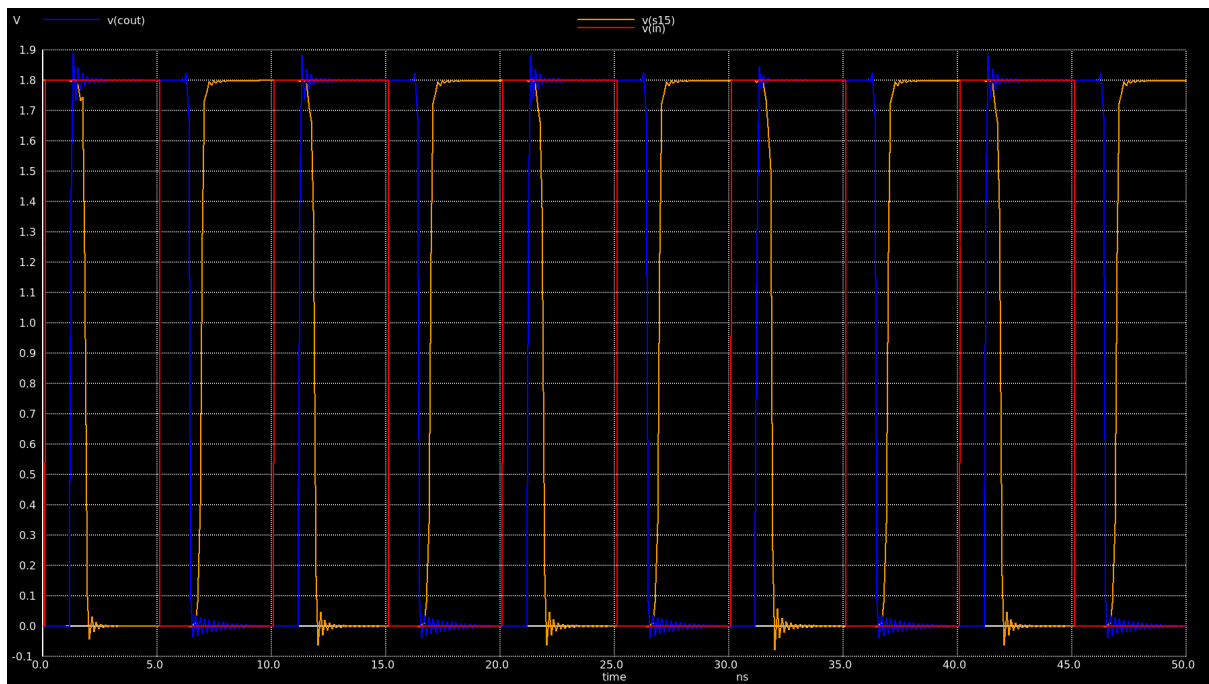


Figure 5: Pre-layout SPICE Simulation Waveforms

```
tb_16bit.spice" -a || sh
x1,x66,net3          0.367237
x1,x66,net4          2.18905e-10
x1,net67             1.8
cout                 5.36616e-07
x1,x68,net1          1.8
x1,x68,net2          1.8
x1,x68,sb            5.36615e-07
x1,x68,net3          0.024894
x1,x68,net4          1.8
x1,x68,net5          2.18908e-10
vin1#branch          0
v5#branch            -1.3229e-08
v1#branch            0

Reference value : 4.85729e-08
No. of Data Rows : 625
tcarry               = 1.092598e-09 targ= 1.195098e-09 trig= 1.025000e-10
thl                  = 1.759554e-09 targ= 1.862054e-09 trig= 1.025000e-10
tlh                  = 1.821140e-09 targ= 6.928640e-09 trig= 5.107500e-09
S15 Delay : 1.79035E-09
Carry Delay : 1.0926E-09
Error: no block to end.
Warning: ignoring previous error
ngspice 10 -> █
```

Figure 6: Pre-layout Delays

Analysis: The sum delay of 1.79035ns represents the critical path delay in the pre-layout simulation. The carry delay is shorter at 1.0926ns due to the bypass mechanism.

5. Layout Steps and Results

5.1 16-bit Carry Bypass Adder Layout

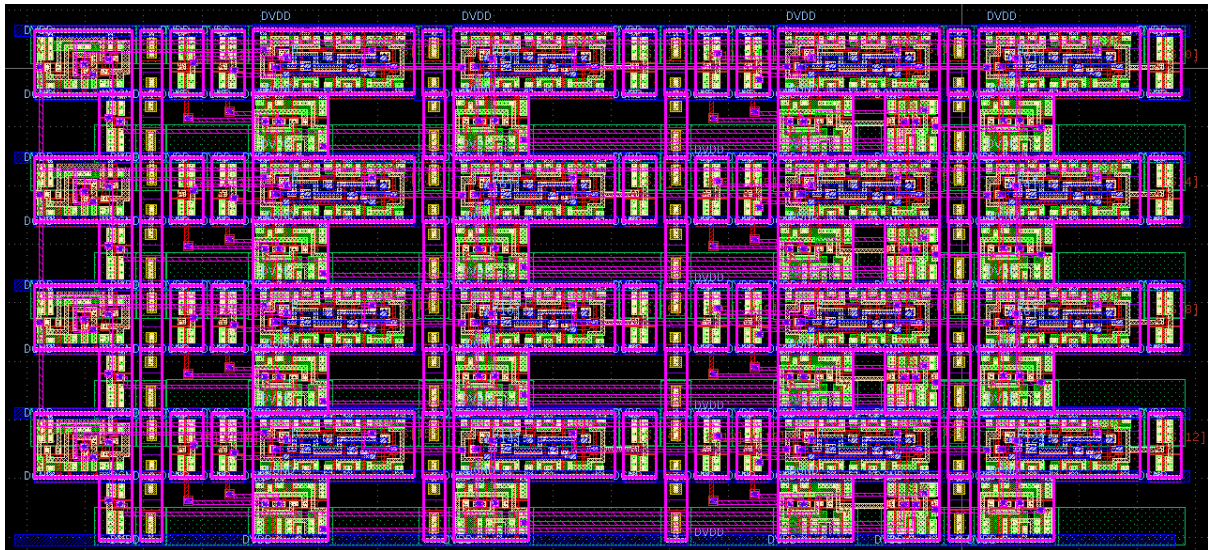


Figure 7: 16-bit Carry Bypass Adder Layout

5.2 DRC Verification

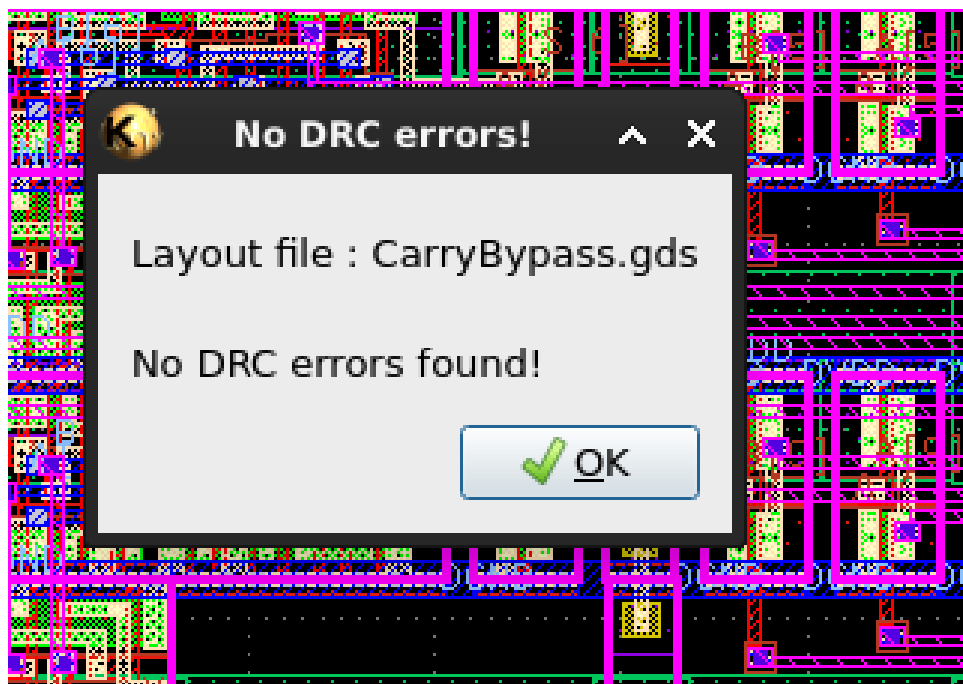


Figure 8: DRC Check Results for 16-bit Carry Bypass Adder - No Errors

5.3 LVS Verification

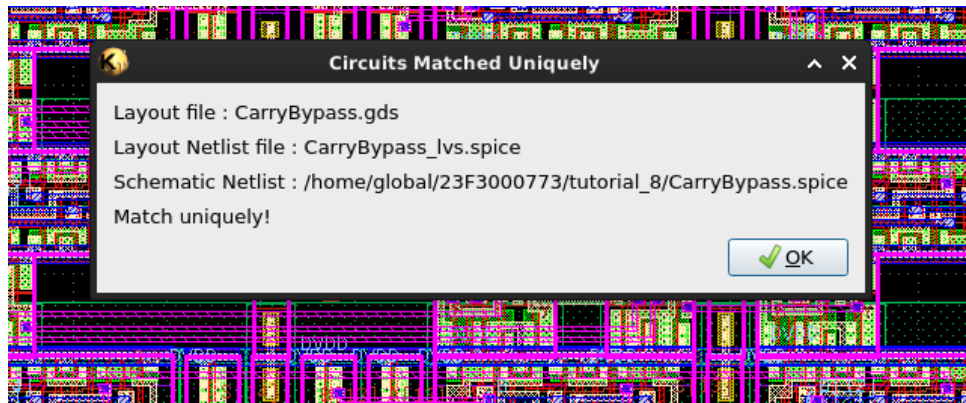


Figure 9: LVS Check Results for 16-bit Carry Bypass Adder - Circuits Matched

6. Post-layout Measurements

6.1 Post-layout Extracted Simulation

The post-layout simulation includes parasitic extraction effects from the physical layout.

Listing 2: Post-layout SPICE Simulation

```
.include CarryBypass_extracted.spice
.control
tran 0.1n 50n
plot v(in) v(Cout) v(S15)
meas tran tcarry trig v(in) val = 0.9 rise=1 targ v(Cout) val=0.9 rise=1
meas tran thl trig v(in) val = 0.9 rise=1 targ v(S15) val=0.9 fall=1
meas tran tlh trig v(in) val = 0.9 fall=1 targ v(S15) val=0.9 rise=1
let delay = ($&thl + $&tlh) /2
echo S15 Delay : $&delay
echo Carry Delay : $&tcarry
end
.endc
```

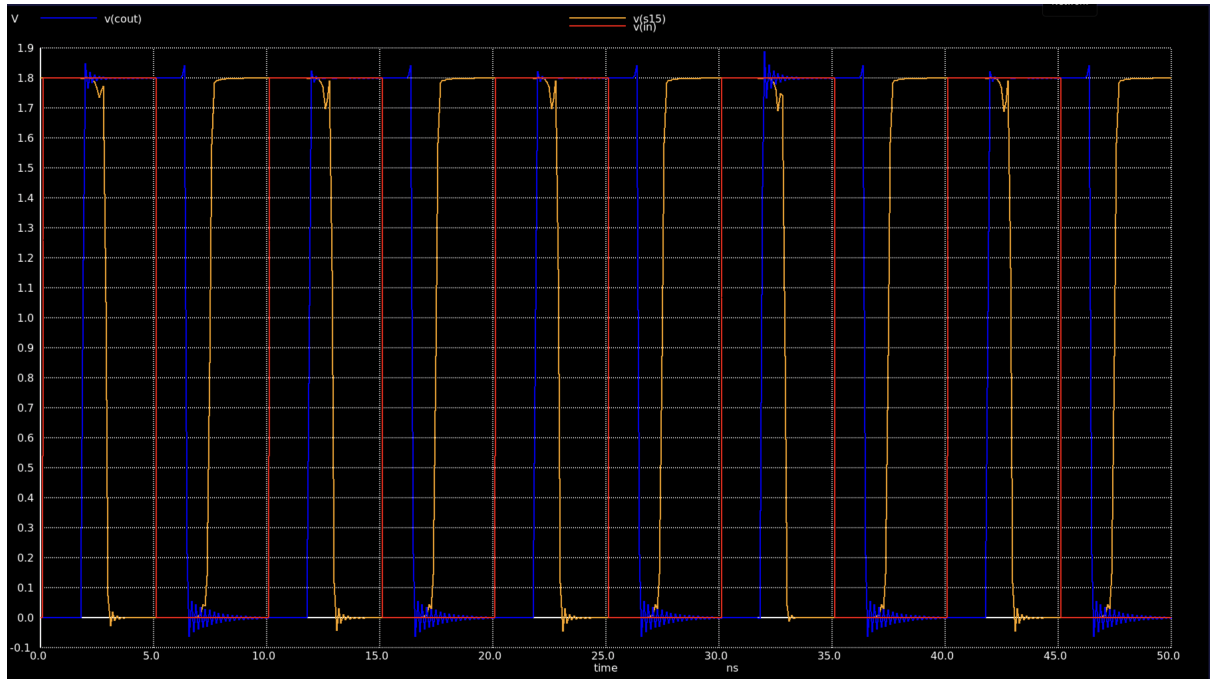


Figure 10: Post-layout Extracted Simulation Waveforms


```

net15                                1.8
net10                                1.8
cout                                6.20728e-07
net12                                1.8
x1.s[11].n0                          1.8
x1.s[11].n1                          1.8
net1                                 1.8
net6                                 1.8
vin1#branch                          0
v5#branch                           -1.20139e-08
v1#branch                           2.76862e-14

Reference value : 4.98634e-08
No. of Data Rows : 626
tcarry      = 1.749761e-09 targ= 1.852261e-09 trig= 1.025000e-10
thl         = 2.790043e-09 targ= 2.892543e-09 trig= 1.025000e-10
tlh         = 2.362675e-09 targ= 7.470175e-09 trig= 5.107500e-09
S15 Delay : 2.57636E-09
Carry Delay : 1.74976E-09
Error: no block to end.
Warning: ignoring previous error
ngspice 10 ->
x0 = 4.19118e-08, y0 = 1.95111
█

```

Figure 11: Post-layout Delays

Analysis:

- Post-layout critical path delay increased to 2.57636ns.
- This increase is due to parasitic resistance and capacitance from interconnects
- The sum path remains the critical path in both pre and post-layout simulations
- The 4-bit carry bypass structure effectively reduces carry propagation delay

Conclusion

This experiment successfully implemented and verified a 16-bit carry bypass adder in both schematic and layout forms. The key findings include:

- The carry bypass architecture effectively reduces carry propagation delay
- Critical path is dominated by sum generation rather than carry propagation
- Both DRC and LVS verification passed successfully
- The bypass mechanism provides significant performance improvement over ripple carry adders

The implementation demonstrates the trade-offs between area overhead and performance improvement in carry bypass adder architectures.

File Locations:

- 16-bit bypass adder layout: `/home/global/23F3000773/tutorial8/CarryBypass.gds`